C2_Agosto21_estimadores

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Ejemplos

Sea $Y_1, Y_2, \ldots, Y_n$ una muestra aleatoria con $E(Y_i) = \mu$ y $V(Y_i) = \sigma^2$. Demuestre que

1. $\bar{Y}$ es un estimador insesgado del parámetro μ.
2. S^2 es un estimador insesgado del parámetro σ^2. **¡Ejercicio para el estudiante!**

② $\boxed{\sum (y_i - \bar{y})^2 = \sum_{i=1}^{n} y_i^2 - n\bar{y}^2}$ → lo demostramos la sesión anterior

$$S^2 = \frac{\sum (y_i - \bar{y})^2}{n-1}$$

$$E(S^2) = E\left(\frac{\sum (y_i - \bar{y})^2}{n-1}\right)$$

$$= \frac{1}{n-1} E\left(\sum_{i=1}^{n} (y_i - \bar{y})^2\right)$$

$$= \frac{1}{n-1} E\left(\sum_{i=1}^{n} y_i^2 - n\bar{y}^2\right)$$

$$= \frac{1}{n-1} \left[E\left(\sum_{i=1}^{n} y_i^2\right) - E(n\bar{y}^2)\right]$$

$$= \frac{1}{n-1} \left[\sum_{i=1}^{n} \underbrace{E(y_i^2)} - n E(\bar{y}^2)\right]$$

$$= \frac{1}{n-1} \left[\sum_{i=1}^{n} (\sigma^2 + \mu^2) - n(\sigma^2_{\bar{y}} + \mu^2)\right]$$

$$= \frac{1}{n-1} \left[n\sigma^2 + n\mu^2 - n\left(\frac{\sigma^2}{n} + \mu^2\right)\right]$$

$$= \frac{1}{n-1} \left[n\sigma^2 + \cancel{n\mu^2} - \sigma^2 - \cancel{n\mu^2}\right]$$

$$= \frac{1}{n-1} \left[n\sigma^2 - \sigma^2\right]$$

$$= \frac{1}{\cancel{n-1}} \sigma^2 \cancel{(n-1)}$$

$$= \sigma^2$$

☑

$$\left\{ \begin{aligned} \sigma^2 &= var(y) \\ &= E(y_i^2) - E(y_i)^2 \\ &= E(y_i^2) - \mu^2 \end{aligned} \right.$$

$$\boxed{\sigma^2 + \mu^2 = E(y_i^2)}$$

$$var(\bar{y}) = \sigma^2_{\bar{y}}$$

$$E(\bar{y}^2) - E(\bar{y})^2 = \sigma^2_{\bar{y}}$$

$$E(\bar{y}^2) - \mu^2 = \sigma^2_{\bar{y}}$$

$$E(\bar{y}^2) = \sigma^2_{\bar{y}} + \mu^2$$

$$\boxed{E(\bar{y}^2) = \frac{\sigma^2}{n} + \mu^2}$$

$$E(\hat{\theta}) = a\theta + b \quad , \quad a \neq 0, \ b \neq 0$$

①
$$B(\hat{\theta}) = E(\hat{\theta}) - \theta$$
$$= a\theta + b - \theta$$
$$= \theta(a-1) + b$$

②
$$E(\hat{\theta}) = a\theta + b$$

$$E(\hat{\theta}) - b = a\theta$$

$$\frac{E(\hat{\theta}) - b}{a} = \theta$$

$$E\Big(\underbrace{\frac{\hat{\theta} - b}{a}}_{\hat{\theta}^*}\Big) = \theta$$

$$\boxed{\hat{\theta}^* = \frac{\hat{\theta} - b}{a}}$$

③
$$MSE(\hat{\theta}^*) = V(\hat{\theta}^*) + (B(\hat{\theta}^*))^2$$
$$= V(\hat{\theta}^*) + 0^2$$
$$= V(\hat{\theta}^*)$$
$$= V\left(\frac{\hat{\theta} - b}{a}\right)$$
$$= \frac{1}{a^2} V(\hat{\theta} - b)$$
$$= \frac{1}{a^2} V(\hat{\theta})$$

$$\Big\{ \quad V(\alpha X) = \alpha^2 V(X)$$

$n = 1000 \quad , \quad y = 560$

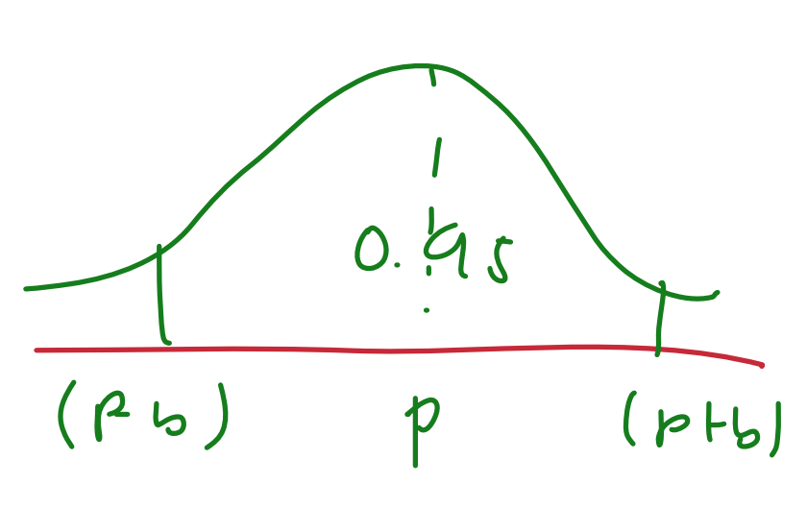
$$\hat{p} = \frac{y}{n} = \frac{560}{1000} = 0.56$$

$$SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}} \approx \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{(0.56)(0.44)}{1000}} = 0.0157$$

$$b = 2SE(\hat{p}) = 2(0.0157) = 0.03139$$

$$\varepsilon = |\hat{\theta} - \theta| = |\hat{p} - p|$$

$$P(\varepsilon < b) = P(|\hat{p} - p| < 0.03)$$
$$= 0.95$$



La probabilidad de que la distancia entre $\hat{p}$ y el verdadero valor p, sea máximo de 0.03, es de 95%, con $\hat{p} = 0.56$.

$\hat{\theta}_1 := \text{mediana}$

$\hat{\theta}_2 := \text{media muestral}$

$$eff(\hat{\theta}_1, \hat{\theta}_2) = \frac{V(\hat{\theta}_2)}{V(\hat{\theta}_1)}$$

$$= \frac{\sigma^2/n}{(1.2533)^2(\sigma^2/n)}$$

$$= \frac{1}{(1.2533)^2}$$

$$= 0.6366 < 1.$$

Por tanto nos quedamos con $\hat{\theta} = \bar{x}$.

$$\frac{V(\hat{\theta}_2)}{V(\hat{\theta}_1)} = 0.6366$$

$$V(\hat{\theta}_2) = 0.6366\, V(\hat{\theta}_1)$$

$\bar{y}$

La varianza de media muestral, es 0.6366 de la varianza de la mediana